

Temporal and Multimodal Visual Intelligence

Does a sequence of observations carry information that a single frame does not?

Technical report | ML Research Engineer take-home assessment

Key findings

- **One frame is not enough for direction.** A single center frame scores 55.81% overall but only 46.3% on the four directional classes, against 94.25% on the Tearing control.
- **Order, not just extra frames, drives the gain.** An ordered Transformer reaches 77.22% and a GRU 75.40%. The same Transformer without positional encoding sees all eight frames but scores 61.96%: order is worth about 15 points, extra frames alone about 6.
- **Shuffling the test frames removes the gain.** Both ordered models fall by about 20 points to 55 to 56%. The order-free model does not move, which confirms the test changes only the order.
- **A zero-shot VLM depends on order too, with caveats.** GPT-4o scores 68% on ordered frames and 44% shuffled over 25 videos. The sample is small and its errors cluster on two direction labels.

1. Question and approach

The question is whether a model can use information that is spread across a sequence and missing from any single frame. I chose a task where that is true by construction and added a control where it should not be. Four classes from Something-Something V2 are direction pairs (pushing left to right and right to left, moving up and down). A fifth, Tearing something into two pieces, shows its outcome in the final frames and serves as the control.

All models read the same frozen DINOv2-small embeddings, so they differ only in how many frames they see and whether they know the order. That gives a three-step ladder: one frame, eight frames without order, eight frames with order. Each step isolates one possible source of improvement. A GPT-4o experiment then asks the same question of a vision-language model without any training.

2. Dataset and sequence analysis

The subset has 4,763 training and 439 validation videos with roughly balanced classes (Figure 1; the validation set holds 73 to 97 videos per class). Each video is reduced to 8 evenly spaced frames at 224x224. Ordering was verified by checking that the extracted frame indices increase strictly for every video, so frame k is always later than frame $k-1$.

What changes, and when. Figure 2 shows one strip per class. In the pushing and moving clips the object sits still for the first few frames, the hand enters around the middle, and the displacement that separates left from right or up from down appears in the second half. In Tearing the two separated pieces are visible at the end, so one late frame is often enough and order should matter little.

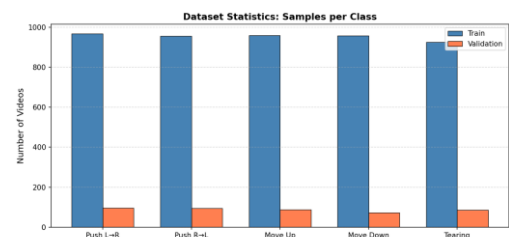


Figure 1: Videos per class in the training and validation splits.

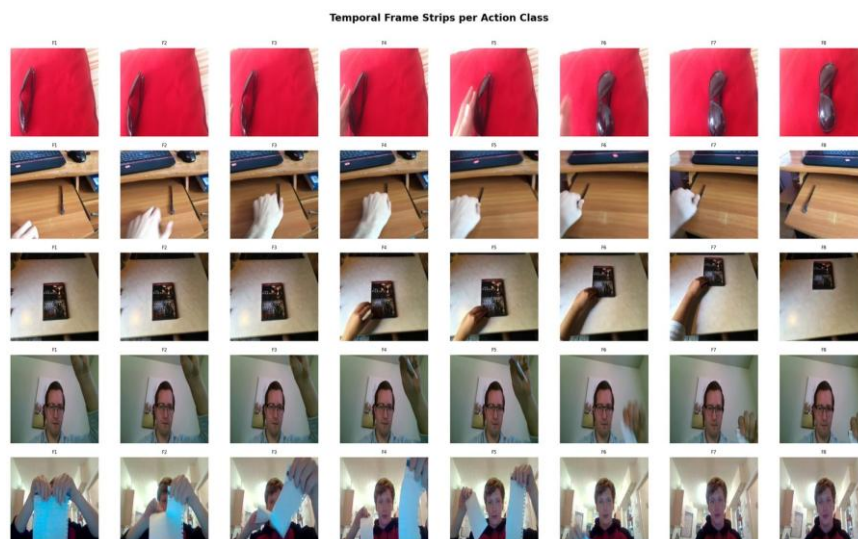


Figure 2: Temporal frame strips, 8 evenly spaced frames per class. Mirror classes look alike in any single early frame.

3. Frame embeddings

Each frame goes through frozen DINOv2-small and the 384-dimensional CLS token is kept, giving an (8, 384) array per video that is cached to disk. All later experiments train on these small arrays with no image loading.

Figure 3 plots the cosine similarity between consecutive frame embeddings. For Push left to right it falls from about 0.94 to about 0.81 by the fifth-to-sixth transition, so most of the change happens in the second half, which is where accuracy starts to climb in Section 5.3. Tearing changes more between frames (similarity mostly 0.70 to 0.81) yet needs order least. So the size of the embedding change shows that something moves, not whether the order of the frames matters. Only the classification experiments can answer that.

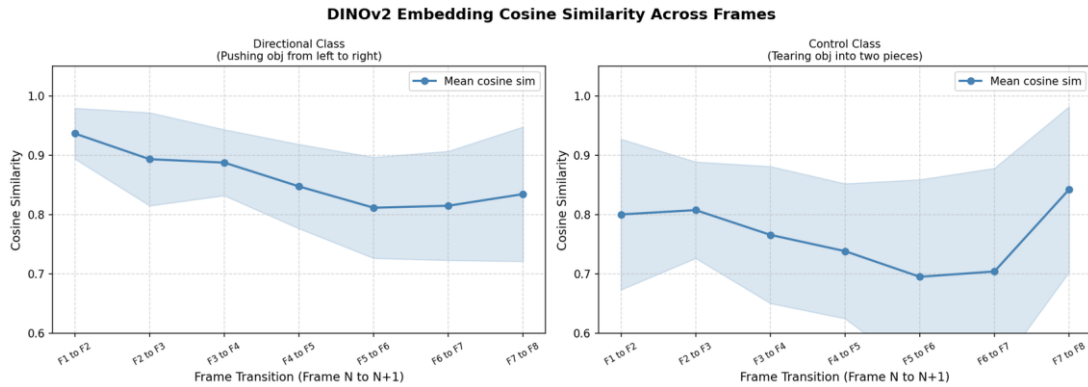


Figure 3: Mean cosine similarity between consecutive DINOv2 frame embeddings; shaded band shows spread across videos.

4. Models and training

Model	Architecture	Input	Can use order?
Single-frame baseline	MLP 384-256-128-5	Center frame (4 of 8)	No (one frame)
Temporal GRU	1-layer GRU, hidden 256, MLP head	8 frames	Yes (recurrence)
Transformer + PE	2 layers, 4 heads, FF 512, sinusoidal PE	8 frames	Yes (positions)
Transformer, no PE	Same encoder, no positional encoding	8 frames	No (permutation invariant)

Table 1: The four models. All share the same frozen embeddings, split, and schedule (40 epochs, learning rate 1e-3, linear warmup then cosine decay).

The no-PE Transformer is the key control. Shuffling the test frames of an ordered model is a weak check on its own, because the model then meets sequences unlike its training data. The no-PE model has the same inputs and capacity but cannot use order by construction, so comparing it with the PE model isolates order directly.

5. Experiments and results

5.1 Single frame versus temporal models

Model	Ordered	Shuffled	Change	Directional (352)	Tearing (87)
Single-frame baseline	55.81%	n/a	n/a	46.3%	94.25%
Temporal GRU	75.40%	55.13%	-20.27	69.6%	98.85%
Transformer + PE	77.22%	56.26%	-20.96	71.6%	100.00%
Transformer, no PE	61.96%	61.96%	0.00	52.8%	98.85%

Table 2: Validation accuracy (439 videos). Directional is the four direction classes pooled, computed from the per-class results.

Both ordered models beat the single frame by about 20 points. A 95% interval on 439 videos is about ± 4.5 points, and the intervals for the single frame (51.1 to 60.4) and the ordered models (71.2 to 80.9) do not overlap. The GRU and the Transformer differ by 8 videos, which is inside that noise, so I do not rank them.

Class (validation videos)	Single frame	GRU	Transformer + PE	Transformer, no PE
Push left to right (97)	51.55%	76.29%	64.95%	59.79%
Push right to left (94)	61.70%	79.79%	77.66%	68.09%
Move up (88)	32.95%	55.68%	72.73%	39.77%
Move down (73)	35.62%	64.38%	71.23%	39.73%
Tear into two pieces (87)	94.25%	98.85%	100.00%	98.85%

Table 3: Per-class validation accuracy. Shading scales with accuracy; bold marks the best model per class.

The gain sits in the directional classes: 46.3% rises to 69.6% (GRU) and 71.6% (Transformer + PE). The vertical classes show it most clearly. A single frame scores 33.0% on Move up and 35.6% on Move down, below the 50% that guessing within an up/down pair would give, so its errors are not confined to that pair. The no-PE Transformer stays at 39.8% and 39.7%. Only the ordered models lift these classes, to between 55.7% and 72.7%. Tearing is already at 94.25% from one frame, and the ordered models add 4 and 5 videos out of 87, which I treat as no change. The GRU looks stronger on Push left to right and the Transformer on the vertical classes, but with 73 to 97 videos per class those gaps are suggestive, not conclusive.

5.2 Order, or just more frames?

Three results answer this together (Figure 4a):

- **Order-free control.** The no-PE Transformer scores 61.96%. That is 6.15 points above the single frame, from seeing more frames, and 15.26 below the PE version. Of the roughly 21-point gain of Transformer + PE over a single frame, about 6 points come from extra frames and about 15 from order.
- **Shuffled test frames.** The GRU drops from 75.40% to 55.13% and Transformer + PE from 77.22% to 56.26%. The no-PE model stays at 61.96%, which confirms the shuffling code only reorders frames.
- **Caveat on shuffling.** The shuffled ordered models land at 55 to 56%, below the 62% of the order-free model, because they receive sequences unlike anything seen in training. Shuffling therefore overstates what order alone is worth. The trained no-PE control, about 15 points, is the fairer number.

5.3 Progressive observation

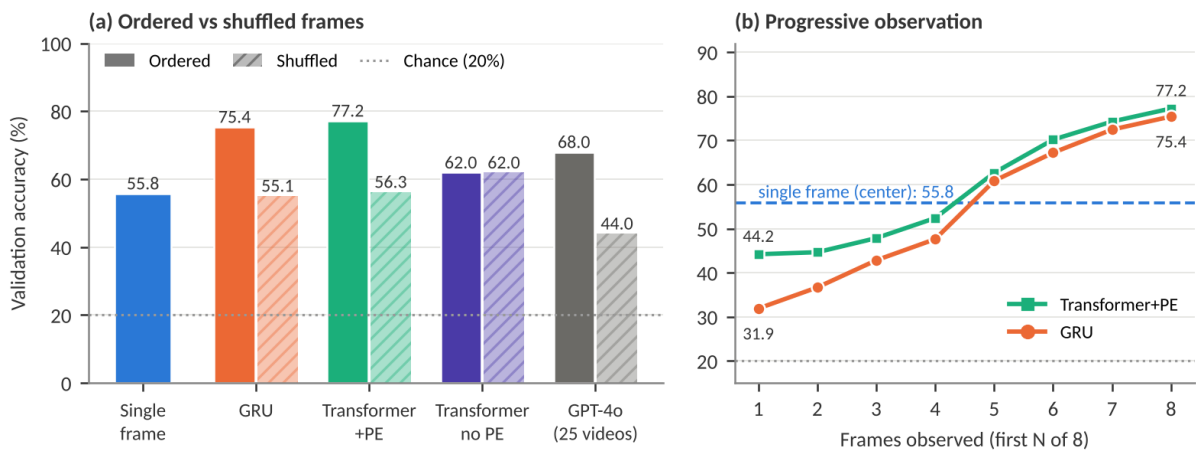


Figure 4: (a) Ordered and shuffled accuracy; GPT-4o is evaluated on 25 videos, the others on 439. (b) Accuracy when only the first N frames are given.

Accuracy rises steadily with the number of frames (Figure 4b): the GRU goes from 31.89% with one frame to 75.40% with eight, and Transformer + PE from 44.19% to 77.22%. Both pass the single-frame level between frames 4 and 5 (GRU 47.61% to 60.82%, its largest jump; Transformer + PE 52.39% to 62.64%), which is when the hand has entered and the object starts to move in Figure 2. Two cautions apply. The models were trained on full 8-frame sequences, so low scores on short prefixes mix missing information with a train-test mismatch. And the prefix starts at frame 1 while the baseline uses the center frame, so the dashed line is a reference, not a like-for-like comparison.

5.4 Training behavior

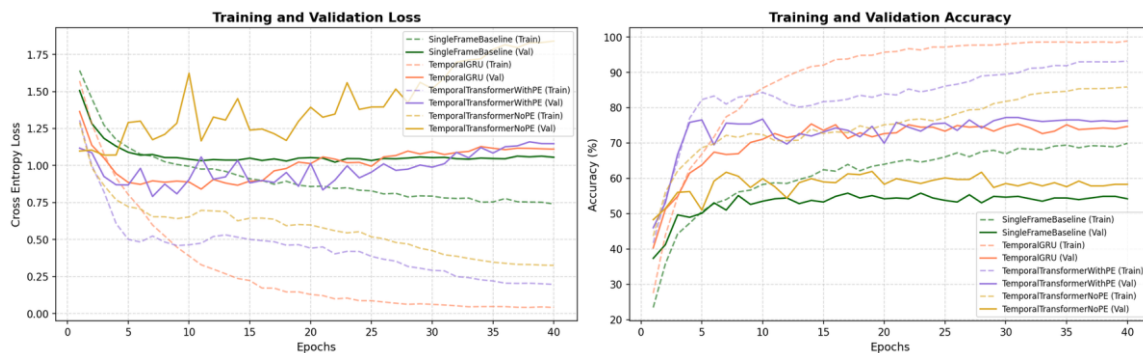


Figure 5: Training and validation loss and accuracy over 40 epochs.

The ordered models fit the training set far better than they generalize. The GRU reaches about 98% training accuracy while validation accuracy sits near 75%, and validation loss for both ordered models bottoms out around epochs 7 to 15 and then rises from roughly 0.85 to about 1.1 while accuracy stays flat. The reported accuracies are from the saved best-validation checkpoint, and the curves end 1 to 4 points lower for some models (the no-PE Transformer finishes near 58%). Section 9 covers what this means for the claims.

5.5 Success and failure cases

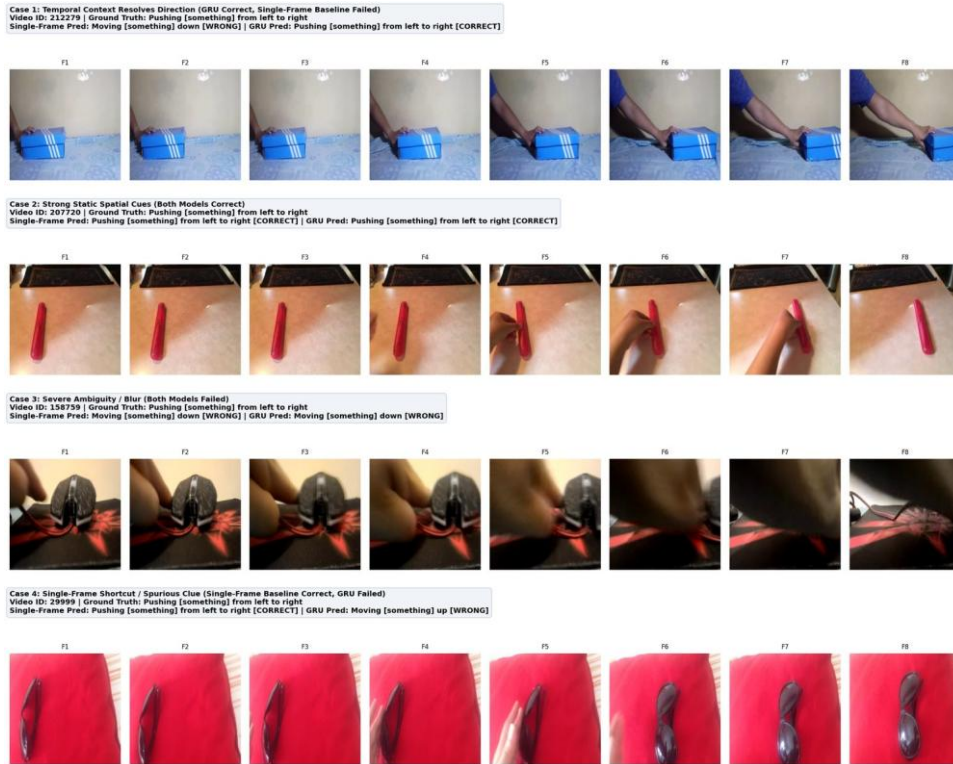


Figure 6: Four representative validation cases found by script; predictions are listed in Table 4.

Case	Video	True label	Single-frame prediction	GRU prediction
1	212279	Push left to right	Move down (wrong)	Push left to right (right)
2	207720	Push left to right	Push left to right (right)	Push left to right (right)
3	158759	Push left to right	Move down (wrong)	Move down (wrong)
4	29999	Push left to right	Push left to right (right)	Move up (wrong)

Table 4: Predictions for the cases in Figure 6.

The cases were selected to illustrate each mode, and all four happen to be Push left to right clips, so they say nothing about how often each mode occurs or about the vertical classes. In Case 1 the hand enters only from frame 4 on, which a single frame cannot see. In Case 2 static cues are enough. In Case 3 camera motion and blur wreck the frames from frame 5, and a frozen per-frame embedding has little left to track. In Case 4 the sunglasses turn from edge-on to flat as they are pushed, which probably reads as vertical motion, though I did not verify this.

6. Vision-language model: GPT-4o

Setup. GPT-4o (OpenAI API, zero-shot) received 4 frames at 336x336 for each of 25 validation videos, 5 per class. Each video was classified twice, once in chronological order and once shuffled. A chain-of-thought prompt asked the model to describe the motion, name the frame with the most decisive directional evidence, and choose one of the five labels.

Class (5 videos each)	Ordered	Shuffled	Change
Push left to right	40%	20%	-20 pp
Push right to left	100%	60%	-40 pp
Move up	20%	0%	-20 pp
Move down	100%	60%	-40 pp
Tear into two pieces	80%	80%	0
All 25 videos	68% (17/25)	44% (11/25)	-24 pp

Table 5: GPT-4o zero-shot accuracy with ordered and shuffled frames.

Order matters to it. 17 of 25 videos are right when ordered and 11 of 25 when shuffled. With 25 videos the 95% intervals (48 to 83% and 27 to 63%) overlap, so this is suggestive rather than established. The size of the drop is similar to the trained models (-20 and -21 points).

The errors look like a label bias. Ordered accuracy is 100% on Push right to left and Move down but 40% and 20% on their mirrors. Shuffled accuracy stays at 60% on those same two classes and falls to 20% and 0% on the mirrors. That fits a prior toward “right to left” and “down” better than it fits motion tracking. The two rationales I inspected agree. On a left-to-right clip

(video 85446) the model described motion from right to left when ordered and downward when shuffled. On a right-to-left clip (163169) it was right both times, and its shuffled rationale still treated Frame 1 as the starting position, so that success says little about order. Each class cell holds only 5 videos, so a confusion matrix over the logged predictions is needed to confirm the bias.

Where the VLM adds value. GPT-4o reaches 68% with no training, against 77.22% for the trained Transformer, but it saw 4 frames of 25 videos while the trained models saw 8 frames of 439, so the comparison is only indicative. What it offers that the embedding models do not: no training data, an open label set, a readable rationale and a named decisive frame, which help with debugging and with flagging ambiguous clips. What it lacks is reliable direction, which is the hard part here, and its output is costly and not deterministic. The rationales are generated text, not evidence of what the model computed. In the system design the VLM therefore sits after the trained temporal model to explain and flag, not to decide direction.

7. Research analysis

A. When did temporal information help? On the four directional classes, where a single frame cannot tell mirror actions apart: 46.3% to 69.6% or 71.6% pooled, and 33 to 36% to 56 to 73% on the vertical classes. The benefit appears only once the motion has started, from about frame 5.

B. When did it not help? On Tearing, which a single frame already gets at 94.25% (the ordered models add 4 or 5 of 87 videos). Also on clips where blur destroys the frames (Case 3), and in at least one clip where the GRU did worse than the single frame (Case 4). Extra frames without order barely help the vertical classes (about 40%).

C. How do we know the model uses temporal relationships and not just more frames? By holding the information fixed and varying only order. The no-PE Transformer sees the same eight frames with the same capacity and scores 61.96% against 77.22%. Shuffling the test frames (–20 points) and progressive prefixes (Figure 4b) support this, with the caveat in Section 5.2 that shuffling overstates the effect. The sharpest test I have not run is time reversal: a model that tracks motion should swap its left-to-right and right-to-left predictions (and up and down) when the clip is played backwards.

D. What result would show the temporal approach is not useful? The no-PE model matching the PE model within noise (about ± 4.5 points), shuffled accuracy staying near ordered accuracy, a flat progressive curve, or a gain that vanishes across seeds or on a held-out test set. None of these holds here, but every model has one run.

E. What would give stronger evidence? Several seeds with intervals, a separate test split, and confusion matrices instead of per-class accuracy alone. Training on shuffled and on reversed sequences. Corruption experiments (blur, dropped frames) comparing single-frame and temporal models. More classes and a second dataset, with non-directional controls. A motion-aware baseline such as optical flow, to check whether frozen CLS embeddings are discarding motion. A larger VLM sample with a video-capable model.

8. System design

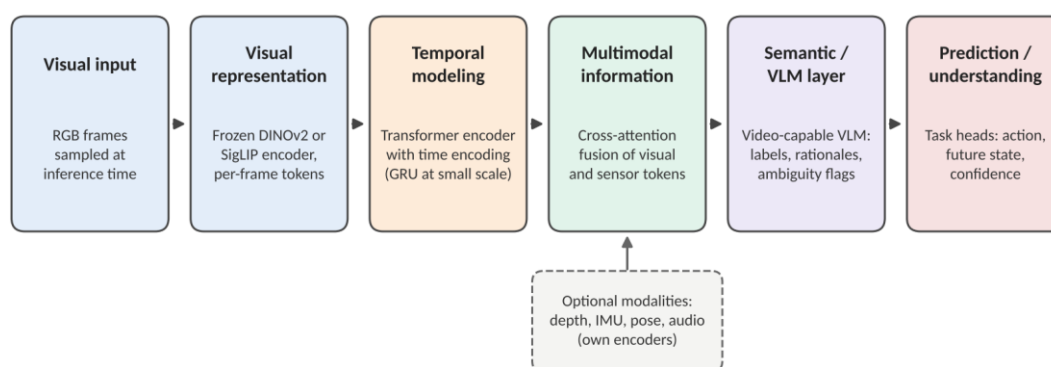


Figure 7: Proposed evolution of this experiment into a multimodal temporal perception system.

A frozen pretrained encoder keeps training cheap and is what this study validated; I would move from CLS tokens to patch tokens so spatial position survives. A Transformer with time encoding replaces the GRU once clips are longer and data is larger, and its attention maps show which frames matter. Depth, IMU, pose and audio get their own encoders and join through cross-attention, which tolerates a missing modality. A video-capable VLM sits after fusion, taking the temporal model's prediction and confidence and producing labels, rationales and ambiguity flags, while the trained model owns direction because it was more reliable here. Lightweight task heads (action, future state, confidence) share the backbone.

9. Limitations and future improvements

- **Single runs, no test split.** Each model was trained once, and checkpoints were selected on the same 439 validation videos used for reporting. The figures are mildly optimistic (Figure 5 ends 1 to 4 points lower), and gaps under about 4.5 points are not resolved. The 20-point effects are far larger than that.
- **Narrow task.** Five classes, one control, one dataset, and 8 sparse frames per clip.
- **Frozen small encoder with CLS tokens only.** Fine motion may be lost, which fits the blur failure.
- **VLM test is small.** 25 videos, 4 frames, one prompt, one model, and no confusion matrix.
- **Not run:** corruption and occlusion experiments and viewpoint analysis, which the brief lists as options. The controls and extensions in 7E are the next steps, together with temporal attention visualization.

10. Conclusion

On this task the answer is yes: a sequence carries information that no single frame does, and most of it is about order. A matched order-free control puts the value of order at about 15 points, extra frames alone at about 6, and the gain is concentrated in the directional classes while the Tearing control shows where order adds nothing. A zero-shot VLM shows a similar dependence on order but is not reliable on direction. The main caveat is that every number comes from one run on 439 validation videos that also served for checkpoint selection, so the next step is to repeat the key comparisons with several seeds and a held-out test set.

Third-party resources: Something-Something V2 (Kaggle mirror), DINOv2 (Meta AI), GPT-4o (OpenAI API), PyTorch.